CMU-Africa MSIT + Mastercard Foundation Scholars: Submission Pack

Kigali, Rwanda (African opportunity). Early deadline 15 Dec 2026: submit by 12 Dec

Evidence status:

- africa.engineering.cmu.edu cannot be opened from this session, so every CMU-Africa fact below is **search-indexed (E2)**. Confirm each in your browser before submitting.
- Mastercard Foundation facts are **verified on the official site (E1)**.

Nothing is submitted, signed or sent for you.

1. Official links

Purpose	URL	Evidence
Admissions (start here)	https://www.africa.engineering.cmu.edu/admissions/index.html	E2
Step 1: online application	https://www.africa.engineering.cmu.edu/admissions/how-to-apply/step-1.html	E2
FAQ (components, tests, fees)	https://www.africa.engineering.cmu.edu/admissions/faq.html	E2
Transcript requirements	https://www.africa.engineering.cmu.edu/admissions/how-to-apply/transcript-requirements.html	E2
Tuition and financial aid	https://www.africa.engineering.cmu.edu/admissions/tuition.html	E2
Mastercard Foundation Scholars at CMU-Africa	https://www.africa.engineering.cmu.edu/impact/mastercard-foundation-scholars.html	E2
Mastercard Foundation partner page	https://mastercardfdn.org/en/what-we-do/our-programs/mastercard-foundation-scholars-program/where-to-apply/carnegie-mellon-university-africa-cmu-africa/	**E1**
Admissions e-mail	applyafrica@andrew.cmu.edu	E2

2. Your position (honest)

Decision	Requirement	You	Status
Admission (MSIT)	Bachelor’s degree; no GPA cut-off, but “most students have graduated with a GPA of 3.0 or above”; strong programming; English	BSc CS, CGPA 2.62/4.00 (as printed); final year 3.20 (own calculation); ~3.5 years in fintech infrastructure (HR letters pending)	Eligible to apply; GPA is the weak point
Need-based financial aid	Financial need, form after you apply; aid concerns admitted African nationals	Ethiopian	Depends on your real figures

Decision	Requirement	You	Status
Mastercard Foundation Scholar	Admitted first; academic talent, giving back, leadership; “highest social and economic barriers” (E1)	Service-oriented work; giving-back evidence to confirm	Possible

3. Portal field map: every field and its status

Portal section	Field	Value / source	Status
Account	E-mail, password	Your private e-mail	**YOU**
Personal	Name (as passport), date of birth, gender, citizenship, residence, address, phone	Passport (private)	**YOU**
Programme	MS in Information Technology, Fall 2027, Kigali	–	READY
Education	Debre Tabor University, BSc Computer Science (Extension), 2017/18–2021/22, awarded 14 Jul 2022	`01_MY_PROFILE`	READY
Education	GPA **2.62** on scale **4.00**, exactly as printed; do not convert or round	Transcript	READY
Transcript upload	Unofficial transcript + grading key, **all years in one PDF**	Your scan	**YOU (scan)**
Employment	SantimPay and Kifiya: titles, dates, duties	HR letters	**BLOCKED (HR letters)**
English test	Leave blank if you apply by 15 Dec (optional); the Duolingo test fee is covered for first-time early applicants	–	READY
Recommenders (3)	Name, title, organisation, e-mail, relationship: (1) final-project supervisor, (2) A− course lecturer, (3) SantimPay manager	Their agreement	**YOU (ask)**
CV	PDF, **max 2 pages**	`06_FINAL_CV/CV_CMU-Africa_2pages`	**DRAFT (needs HR facts)**
Written essays	3 questions + 1 optional, max 300 words each	Section 4	**DRAFT: paste the real questions**
Video essay	Random question, about 30 s to prepare, up to 180 s to answer, no re-recording (E2/E3)	Section 5	**PRACTISE**
Mastercard Foundation interest	Tick yes if asked	–	READY
Declaration + submit	Your own decision	–	**YOU**

Portal section	Field	Value / source	Status
After submission	Financial aid form; technical test invitation	Sections 6, 7	Later

4. Written essays: final drafts (max 300 words each)

Important:

- CMU-Africa states that plagiarism leads to denial. These are your drafts built from your facts.
- **Rewrite each one in your own voice before submitting**, then fill the few `[…]` gaps that only HR letters can supply.
- Paste each real portal question above the closest essay.

Essay A: Why the MSIT at CMU-Africa, and your career goals (about 270 words)

At SantimPay in Addis Ababa I work on the infrastructure behind digital payments: card and QR payments in shops, school-fee payments, and payment services that partner banks offer their customers. As [title from HR letter] I [two duties from the HR letter, e.g. operate the Linux servers that run our payment services and maintain their deployment pipelines]. When something fails, a merchant cannot take a payment. Keeping these systems reliable has taught me how much depends on infrastructure that most people never see.

It has also shown me the limits of what I know. In [month, year], [one real incident: what failed, how you found the cause, what changed afterwards]. I could fix the immediate problem, but I lacked the depth in distributed systems, security engineering and system design to make sure such failures do not recur. That is the gap I want the MSIT to close.

CMU-Africa combines Carnegie Mellon's engineering rigour with the realities of African markets. I want to learn [one or two MSIT courses or concentrations from the official curriculum, and why each matters to your work] alongside engineers from across the continent who face the same constraints I do: unreliable connectivity, fast-growing transaction volumes, and the need for security that regulators and customers trust.

My academic record shows steady improvement. My final year was my strongest (GPA 3.20, my calculation from the transcript), with A− grades in Advanced Programming, Compiler Design, Complexity Theory and both parts of my final-year project.

After the MSIT I intend to lead reliability and security engineering for payment platforms in Ethiopia, designing infrastructure that millions of first-time digital users can depend on.

Essay B: Leadership and professional journey (about 250 words)

I graduated in Computer Science from Debre Tabor University in July 2022 and joined SantimPay in [month, year]. [One sentence on later roles and Kifiya, exactly as in the HR letters.]

The clearest test of my leadership came when [one real situation, e.g. a POS rollout for N merchants, a recurring outage, or training new colleagues]. I was responsible for [your mandate]. I [two or three concrete actions: what you decided, how you organised people, how you communicated with partners]. As a result, [what changed; give a number only if you can prove it].

I learned that leading in infrastructure work means being calm when systems fail, explaining technical problems in simple terms to people outside the team, and fixing the process, not only the symptom.

Leadership for me also means helping others grow. In 2022 I worked as a co-trainer with Dereja, supporting trainees on the learning platform and preparing weekly progress reports [confirm dates and the number of trainees]. Watching trainees gain confidence showed me the value of patient, practical teaching.

At CMU-Africa I want to develop these skills further through team projects with classmates from different countries, and through the Mastercard Foundation's leadership programme, so that I can lead engineering teams that build trustworthy digital services in Ethiopia.

Essay C: Contribution to your community and giving back (about 230 words)

Ethiopia is moving rapidly from cash to digital payments. For a small merchant, a reliable payment service means more customers and safer income. For a family, it means paying school fees without travelling to a bank. When these systems fail, the people with the least margin for error lose the most, and they lose trust in digital money.

I have spent my career on the infrastructure behind these services, and I want to make it trustworthy at a much larger scale. After the MSIT I will return to Ethiopia's digital-finance sector to design payment infrastructure that is secure, reliable and affordable to run.

Giving back also means sharing knowledge. I will [one specific, realistic commitment, e.g. run a free workshop each semester for Computer Science students at Debre Tabor University on Linux, automation and reliability; mentor two junior engineers each year]. Universities outside Addis Ababa rarely have access to industry practice, and I know from my own studies how much that would have helped me.

The Mastercard Foundation invests in young Africans who will lift others as they rise. I want to be that kind of engineer: one who builds the systems ordinary people depend on, and who helps the next generation build them too.

Essay D (optional): Academic record (about 160 words)

My cumulative GPA of 2.62/4.00 does not show the direction of my studies. I completed my BSc through Debre Tabor University's extension programme over five academic years. My first years were weak: I failed my first attempts at Probability and Statistics and at Discrete Mathematics, and later passed both. As I moved into core computer-science subjects my results rose. My final-year GPA was 3.20 (my calculation from the transcript), with A− grades in Compiler Design, Advanced Programming, Complexity Theory and both parts of my final-year project. Since graduating I have applied that knowledge every day in production systems. I know mathematics is my weaker foundation, and I am addressing it [only a real step: e.g. "by completing an online discrete-mathematics course before enrolment"].

If the portal's questions differ, map them this way:

- "goals / why CMU-Africa" → A
- "leadership / a challenge you overcame" → B
- "community / impact / giving back" → C
- "anything else / circumstances" → D

5. Video essay: random question, about 30 s to prepare, up to 180 s, no re-recording

Answer frame for any question:

1. One-sentence answer.
2. One real example (Situation → what you did → result).
3. Lesson.
4. A link to the MSIT or your goals.

Aim for 90–150 seconds.

Likely question	Your example
Tell us about yourself	Degree → final-year improvement → payments infrastructure → goal
Describe a time you failed	The incident in Essay A

Likely question	Your example
Describe a time you led others	Essay B example
How will you give back?	The workshop/mentoring commitment
Why technology for Africa?	Merchants and families moving to digital payments
A difficult team situation	A disagreement solved by explaining the technical facts calmly [real]
What makes you unique?	You keep live payment systems running for real users

Practise 5 times on your phone with a 30-second timer before you open the real recording.

6. Financial need statement (fill with exact figures; do not exaggerate)

> I earn [ETB amount] per month (net), which supports [who depends on you]. My savings are [amount]. The MSIT tuition for African students in 2026–27 is US$16,000 (CMU-Africa tuition page), plus living costs in Kigali of about [amount from CMU-Africa's estimate]. Together these are [N] times my annual income. Without the Mastercard Foundation Scholarship I cannot enrol. I have also applied to [other scholarships]; none is secured.

7. Technical test (after the application is complete; fee covered)

Content: analytical and quantitative skills and programming logic (E2).

Plan: two weeks, one hour a day.

- Python basics.
- Arrays, strings, recursion.
- Big-O notation.
- Logic and number puzzles.
- Percentages and rates.
- Two timed mixed sets.

8. Outstanding items (what still blocks submission)

#	Item	Owner	By
1	Open the portal, create the account, **paste the real essay questions** here	You	15 Nov
2	SantimPay + Kifiya HR letters (titles, dates), then fill CV and essays	You → HR	25 Nov

#	Item	Owner	By
3	One real incident story + one leadership story (Essays A and B)	You	22 Nov
4	Three recommenders agreed (names, titles, e-mails)	You	1 Dec
5	Transcript scan + grading key as one PDF	You	1 Dec
6	Dereja dates and numbers, and your giving-back commitment	You	1 Dec
7	Financial figures for the aid form	You	after submitting
8	Video-essay practice (5 takes)	You	10 Dec

9. Submission guide

1. Admissions page → Step 1 → **Apply** (Chrome or Firefox).
2. Fill the personal and education sections from section 3.
3. Upload the transcript PDF and the 2-page CV PDF.
4. Enter the 3 recommenders.
5. Paste your final essays.
6. Record the video essay (one take).
7. Review every page, then **submit by 12 Dec**.
8. Save the confirmation PDF in `13_FINAL_SUBMISSIONS/CMU-Africa/`.
9. After submission: complete the financial aid form; take the technical test when invited.

10. Final audit checklist

- ☐ Name identical to the passport everywhere
- ☐ GPA 2.62/4.00 as printed
- ☐ Every date and title matches the HR letters
- ☐ CV ≤ 2 pages, PDF, no `[…]` left
- ☐ Each essay ≤ 300 words, in your own words, no `[…]` left
- ☐ Three recommenders informed that CMU-Africa will contact them
- ☐ Transcript: all years, one PDF, grading key included
- ☐ Video practised; quiet room; camera at eye level
- ☐ Submitted on or before 12 Dec (deadline 15 Dec); confirmation saved

11. Interview (if invited)

Question	Outline
Walk me through your background	Degree → final-year improvement → payments infrastructure → goal
Why do you need this degree now?	The design gap you hit (Essay A incident)
Your GPA is below most admits	Upward trend; A− in demanding courses; production responsibility; public code
Leadership / failure / conflict	Your STAR stories (Essays A and B)
How will you give back?	The concrete commitment from Essay C
Explain a tool on your CV	Only tools you can explain in depth
How would you secure a payment API?	Authentication, TLS, rate limiting, logging, least privilege